Huawei Access Node HEB-ACC-40 SOP for Hardware/Environmental Fault ALARM_2987

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2987

Description

The Hebbal Edge Huawei Access Node HEB-ACC-40 has experienced a critical hardware/environmental fault, indicated by the ALARM_2987 alarm code. This fault is likely due to an ASIC memory parity error, line card crash, or power supply redundancy loss. The Huawei Access Node, being a critical component of the access network, relies on the robustness of its hardware and environmental conditions to ensure seamless communication between the core network and customer premises equipment. In the event of a hardware failure, the node's ability to process and forward traffic is compromised, resulting in dropped calls, data packet loss, and overall network instability. The Line Card (LC) and Power Supply (PS) redundancy are designed to provide high availability and automatic switchover in case of a failure, but in this scenario, the redundancy has been compromised, leading to a critical fault. The root cause analysis will focus on identifying the specific hardware or environmental factor contributing to the fault, with the goal of restoring the node to a stable and operational state.

Pre-requisite Checks

- Pre-requisite: Verify the site location and Node ID to ensure the correct node is being addressed., Description: This is crucial to ensure that the correct node is being diagnosed and resolved.

- Pre-requisite: Check the redundant link status to confirm that it is still active., Description: This will help identify if the redundancy has been compromised, which could be contributing to the fault.

- Pre-requisite: Verify that the remote hands have been dispatched to the Hebbal Edge site., Description: This will ensure that the site is secure and that the remote hands can perform any necessary physical checks or repairs.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Error	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display device, Expected Output: Device information including ASIC memory status and line card status., Description: This command will provide an overview of the node's hardware configuration and help identify any potential issues with the ASIC memory or line card.
```

```
Command: display power-supply, Expected Output: Power supply status including redundancy and switchover information., Description: This command will provide information on the power supply status, including redundancy and switchover information, which can help identify if the power supply redundancy has been lost.

Command: display fault-management alarm, Expected Output: Alarm information including ALARM_2987 details., Description: This command will provide detailed information on the alarm, including the ALARM_2987 code, which can help identify the root cause of the fault.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC or Line Card, Command: upgrade device replace, Description: This command will initiate the replacement of the faulty ASIC or Line Card, which will resolve the fault and restore the node to a stable state.

Step 2: Step: Re-configure Power Supply Redundancy, Command: configure power-supply redundancy, Description: This command will re-configure the power supply redundancy, which will help ensure that the node has a stable and redundant power supply.

Post-Check Verification

```
Command: display device, Expected Output: Device information including ASIC memory status and line card status., Description: This command will verify that the node's hardware configuration has been restored to a stable state.

Command: display power-supply, Expected Output: Power supply status including redundancy and switchover information., Description: This command will verify that the power supply redundancy has been re-configured and is functioning correctly.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: rollback device configuration, Description: This command will restore the original configuration of the node, which will help ensure that the node returns to its original operational state.